

✓ Portfolio uploaded: NVDA, AAPL, MSFT, TSLA, CASH

EisaX Portfolio Risk Report

Date: March 30, 2026 | Period: 1 Year | Risk-Free Rate: 4.5% (US T-Bill)

Executive Summary

Portfolio returned **+29.1%** vs S&P 500 **+14.1%** (Alpha: **+14.9%**). Risk profile: **Aggressive** ●. Sharpe: **0.85** (rf=4.5%, 1Y Historical). Immediate action: Reduce tech concentration & add defensive assets.

Holdings

Ticker	Weight	Sector	Beta	P/E	Div Yield
NVDA	30.0%	Technology	2.38	34.1	0.02%
AAPL	20.0%	Technology	1.12	31.5	0.41%
MSFT	20.0%	Technology	1.11	22.3	0.95%
TSLA	15.0%	Consumer Cyclical	1.93	332.0	0.00%
CASH	15.0%	—	0	—	0%

Weighted Portfolio Dividend Yield: 0.33%

Risk Metrics

Method: Historical Simulation (252 trading days) · rf = 4.5%

Metric	Value	Assessment
1Y Total Return	+29.1%	● Strong
vs S&P 500 (Alpha)	+14.9%	● Outperforming benchmark
Annualized Volatility	30.6%	● High
Sharpe Ratio (1Y, rf=4.5%)	0.85	● Acceptable
Sortino Ratio (1Y, rf=4.5%)	1.30	● Good downside-adjusted

Metric	Value	Assessment
Portfolio Beta (weighted)	1.70	🔴 High Risk
VaR 95% 1-Day (Historical)	-2.41%	95% of days, loss ≤ this
CVaR 95% (Expected Shortfall)	-3.93%	Avg loss when VaR is breached — tail risk
Max Drawdown (1Y)	-20.0%	Worst peak-to-trough
Max Drawdown Duration	102d	Longest time underwater
Portfolio Div Yield	0.33%	Weighted annual income
Rolling Sharpe (63d)	—	Current (last 63d): -3.38



Sector Exposure

Sector	Weight	HHI Contribution
Technology	82.4%	0.678
Consumer Cyclical	17.6%	0.031

Sector HHI: 0.709 — 🔴 Highly Concentrated (0=perfect diversification, 1=single sector)



Correlation & Diversification

	NVDA	AAPL	MSFT	TSLA
NVDA	1.00	0.52	0.56	0.57
AAPL	0.52	1.00	0.39	0.52
MSFT	0.56	0.39	1.00	0.43
TSLA	0.57	0.52	0.43	1.00

Metric	Value	Interpretation
Avg Pairwise Correlation	0.50	🟡 Moderate co-movement between holdings
Effective N (eigenvalue)	2.3	🟡 Moderate — true independent bets in portfolio

⚠️ **Note:** Even without pairs >0.70, Effective N = 2.3 indicates portfolio behaves like **2.3 independent assets**

— not 4. Sector concentration (not just pairwise correlation) is the real risk driver here.

Stress Testing

Historical crises use actual observed returns + sector-adjusted amplification (NOT linear beta)

Scenario	SPX Actual	Portfolio Est.	Notes
🔴 2022 Rate Shock (actual)	-19.5%	-28.3%	Tech -33% vs SPX -19.5% in 2022
🔴 COVID Crash Mar 2020	-34.0%	-40.8%	SPX -34% in 33 days
🔴 2008 GFC (Sep–Nov)	-42.0%	-37.8%	Financials led; tech -40%
🔴 Bear Market -30% Scenario	-30.0%	-39.0%	Projected; tech typically amplified
🟢 2023 AI Bull Run (actual)	+26.5%	+47.7%	Tech +55% vs SPX +26.5%
🟢 Rate Cut Cycle +15%	+15.0%	+21.0%	Growth stocks benefit most

Tech Weight: 100% of equity. In risk-off events, tech typically amplifies market moves by 1.2–1.8×. Linear beta alone ($\beta=1.70$) understates true drawdown risk.

EisaX Risk Assessment

🔴 **Aggressive** — $\beta=1.70$, CVaR=-3.93%/day, Eff.N=2.3 bets. 82% in Technology. In a 2022-style selloff your portfolio would have lost ~33% vs SPX -19.5%. Diversification is urgently needed.

Performance Attribution (1Y)

Brinson-Hood-Beebower: each holding's contribution to total portfolio return

Ticker	Weight	1Y Return	Contribution	Attribution
NVDA	35.3%	+52.8%	+18.63pp	🟢
TSLA	17.6%	+37.3%	+6.58pp	🟢
AAPL	23.5%	+14.7%	+3.45pp	🟢
MSFT	23.5%	-5.1%	-1.20pp	🔴
CASH/Other	—	—	+1.60pp	⚪
TOTAL	100%	—	+29.06pp	

🏆 **Alpha driver:** NVDA contributed +18.6pp ($+52.8\% \times 35\%$ weight). S&P 500 returned +14.1% — alpha gap of +14.9pp.



Factor Exposure Analysis

Fama-French proxy: factor tilts computed from fundamentals + price momentum

Factor	Portfolio Exposure	Interpretation
Market (Beta)	+1.70	● High market sensitivity
Growth (P/E tilt)	+0.42	● Strong growth tilt — high valuation risk
Size (SMB proxy)	-0.82	● Large-cap dominated (low SMB)
Momentum (12M)	+0.27	● Strong momentum

Factor exposures are approximated from fundamentals. For precise Fama-French loadings, a 3-year return regression against HML/SMB/MKT factors is required.



Institutional Optimization Engine

Convex QP · CLARABEL solver · Zero-tolerance constraint enforcement



Constraint Diagnostics

Reference: \$100,000 portfolio · normalized weights

Constraint	Limit	Current	Status	Required Fix
Single stock: NVDA	$\leq 20\%$	35.3%	● -15.3pp	Sell ~\$15,294 of NVDA
Single stock: AAPL	$\leq 20\%$	23.5%	● -3.5pp	Sell ~\$3,529 of AAPL
Single stock: MSFT	$\leq 20\%$	23.5%	● -3.5pp	Sell ~\$3,529 of MSFT
Sector: Technology	$\leq 40\%$	82.4%	● -42.4pp	Reduce by ~\$42,353 / add non-Technology
Portfolio beta	≤ 1.30	1.70	● +0.40	Add low- β assets: TLT, GLD, BRK-B, XLV



Estimated rebalancing: ~\$22,353 turnover (22.4% of portfolio) to reach basic mandate compliance



Mandate Mode Feasibility

	 Conservative	 Balanced	 Aggressive
Max single stock	10%	15%	20%
Max portfolio β	≤ 1.00	≤ 1.20	≤ 1.50
Max sector weight	30%	35%	40%
Single-stock achievable	✗ Need 10+ holdings	✗ Need 7+ holdings	✗ Need 5+ holdings
Beta achievable	✗ Min $\beta=1.13$	✓	✓
Sector cap achievable	⚠ Current assets only	⚠ Current assets only	⚠ Current assets only

⚠ Constraint issues with current holdings:

- $4 \text{ holdings} \times 10\% = 40\% < 100\%$ (need ≥ 10 holdings for this mandate)
- $4 \text{ holdings} \times 15\% = 60\% < 100\%$ (need ≥ 7 holdings for this mandate)
- $4 \text{ holdings} \times 20\% = 80\% < 100\%$ (need ≥ 5 holdings for this mandate)
- Min achievable $\beta = 1.13$ exceeds cap $\beta \leq 1.00$
- Sector 'Technology': 3/4 holdings $\rightarrow$ max total = $40\% < 100\%$ with current assets
- Sector 'Technology': 3/4 holdings $\rightarrow$ max total = $50\% < 100\%$ with current assets
- Sector 'Technology': 3/4 holdings $\rightarrow$ max total = $60\% < 100\%$ with current assets

💡 To achieve full mandate compliance, add:

- Add 1+ new assets OR use Aggressive mandate ($4 \times 20\% \rightarrow 80\%$ min)
- Add 3+ new assets OR use Aggressive mandate ($4 \times 20\% \rightarrow 80\%$ min)
- Add 6+ new assets OR use Aggressive mandate ($4 \times 20\% \rightarrow 80\%$ min)
- Add diversifying sectors: XLV (Healthcare), TLT (Bonds), GLD (Gold), BRK-B, XLP
- Add: TLT ($\beta \approx -0.5$), GLD ($\beta \approx 0.1$), USMV ($\beta \approx 0.7$), BRK-B ($\beta \approx 0.9$)

Optimal Weight Comparison

Ticker	Current	 Conservative	 Balanced	 Aggressive
NVDA	35.3%	86.8%	1.0%	22.8%
AAPL	23.5%	1.0%	89.2%	64.2%
MSFT	23.5%	1.0%	1.0%	1.0%
TSLA	17.6%	11.2%	8.8%	12.0%
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Ticker	Current	 Conservative	 Balanced	 Aggressive
Sharpe	0.85	1.13	0.54	0.80
Beta	1.70	2.30	1.20	1.50
CVaR/day	-3.93%	-5.10%	-4.39%	-4.24%
Ann. Vol	30.6%	40.0%	31.4%	31.1%
Solver	Baseline	 partial (relaxed: single-stock cap, beta cap, sector cap)	 partial (relaxed: single-stock cap, sector cap)	 partial (relaxed: single-stock cap, sector cap)

Execution Plan

Conservative

Pension / Endowment | Solver:

 partial (relaxed: single-stock cap, beta cap, sector cap)

Action	Ticker	From	To	Δ	\$ on \$100k	Reason
 BUY ↑	NVDA	35.3%	86.8%	+51.5pp	\$51,459	Underweight vs risk-optimal
 SELL ↓	AAPL	23.5%	1.0%	-22.5pp	\$22,529	Exceeds 10% mandate cap
 SELL ↓	MSFT	23.5%	1.0%	-22.5pp	\$22,529	Exceeds 10% mandate cap
 SELL ↓	TSLA	17.6%	11.2%	-6.4pp	\$6,400	Exceeds 10% mandate cap

Multi-Objective Trade-off Analysis

Strategy	Sharpe	CVaR/day	Beta	Ann.Vol	Status
Current portfolio	0.85	-3.93%	1.70	30.6%	Baseline
 Conservative	1.13 (+0.28)	-5.10% (-1.17pp)	2.30 (+0.60)	40.0%	 partial (relaxed: single-stock cap, beta cap, sector cap)
 Balanced	0.54 (-0.31)	-4.39% (-0.46pp)	1.20 (-0.50)	31.4%	 partial (relaxed: single-stock cap, sector cap)
 Aggressive	0.80 (-0.05)	-4.24% (-0.31pp)	1.50 (-0.20)	31.1%	 partial (relaxed: single-stock cap, sector cap)

Post-Rebalance Simulation

Best mandate:

 Conservative · Pension / Endowment

Metric	Before	After	Δ	Verdict
Sharpe Ratio	0.85	1.13	+0.28	✅ Improved
Ann. Volatility	30.6%	40.0%	+9.35	🔴 Higher
Portfolio Beta	1.70	2.30	+0.60	⚠️ Similar
CVaR 95%/day	-3.93%	-5.10%	-1.17	🟡 Similar
Single-stock cap	🔴 Breach	⚠️ Partial	—	⚠️
Beta mandate	🔴 Breach	⚠️ Near limit	—	✅

 Portfolio Assessment

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 Audit Trail

For compliance, reproducibility, and institutional trust

Field	Value
Report ID	89b2dd13-c14a-4b0b-8fce-a7edae2562fe
Generated	2026-03-30 12:52:20 UTC
Price Data As-Of	2026-03-27
Period Analysed	2025-03-28 → 2026-03-27 (252 trading days)
Data Source	Yahoo Finance (yfinance) — 15-min delayed
Tickers Fetched	NVDA, AAPL, MSFT, TSLA
Risk-Free Rate	4.50% (US 3-Month T-Bill)
Methodology	Historical Simulation · Brinson Attribution · Fama-French proxy
Optimizer	CLARABEL convex QP (cvxpy) — strict constraint enforcement
Report Hash	sha256:0800c31e01776ba7...



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